Chromosomal Mutations

- Involve all or part of a chromosome
- Occur during meiosis
- Affect many genes, rather than one
- Several types
  - Deletion: part of chromosome is lost
  - Duplication: part is doubled – occurs twice
  - Inversion: part is flipped
  - Translocation: part breaks off and joins to another chromo
  - Non-disjunction: extra chromo (trisomy), or one less (monosomy). *Aneuploidy* means change in number of chromos.
- Cause (often severe) abnormalities or death in utero

Learning Aim: Define Chromosomal Mutation

Chromosomal Mutation Types

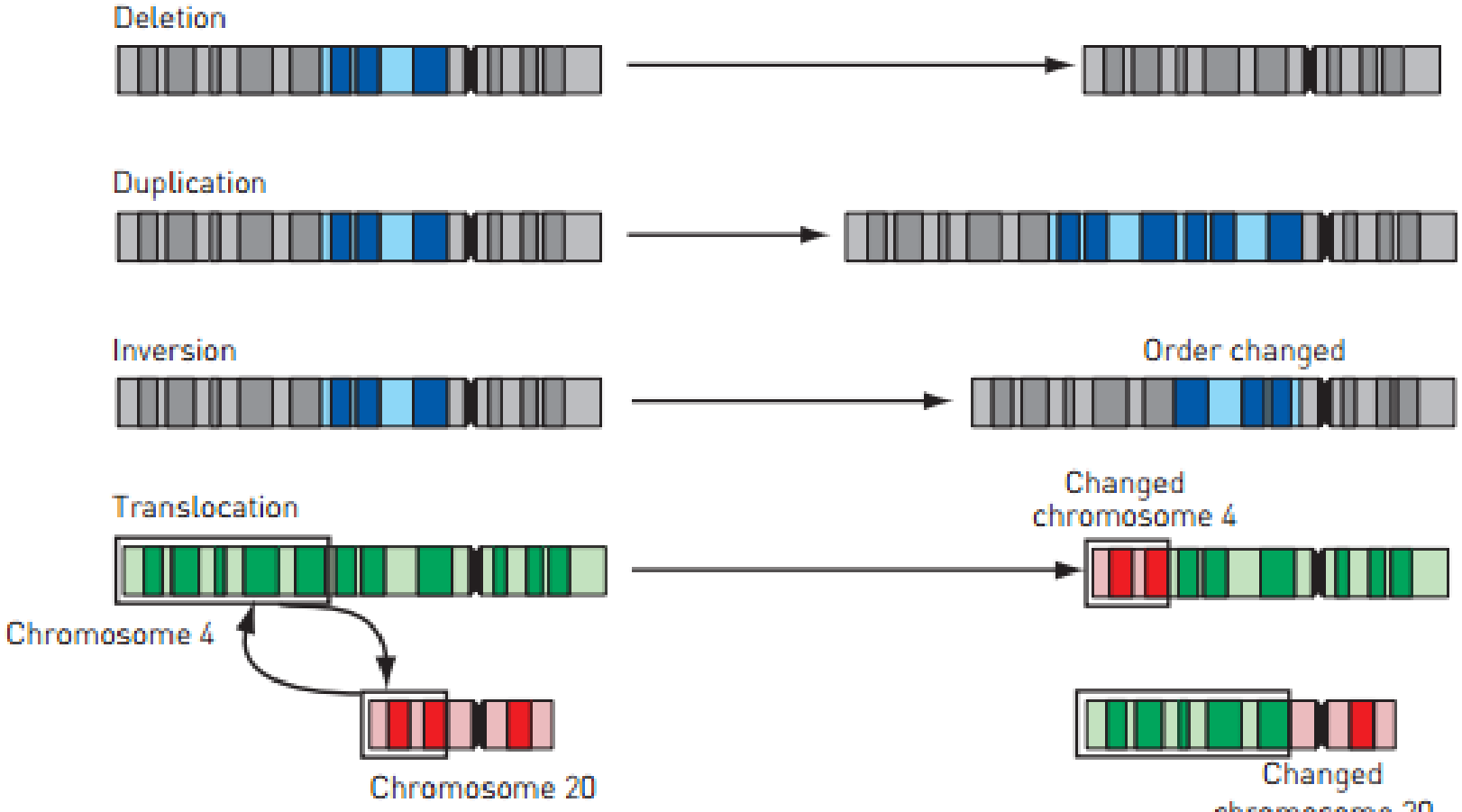


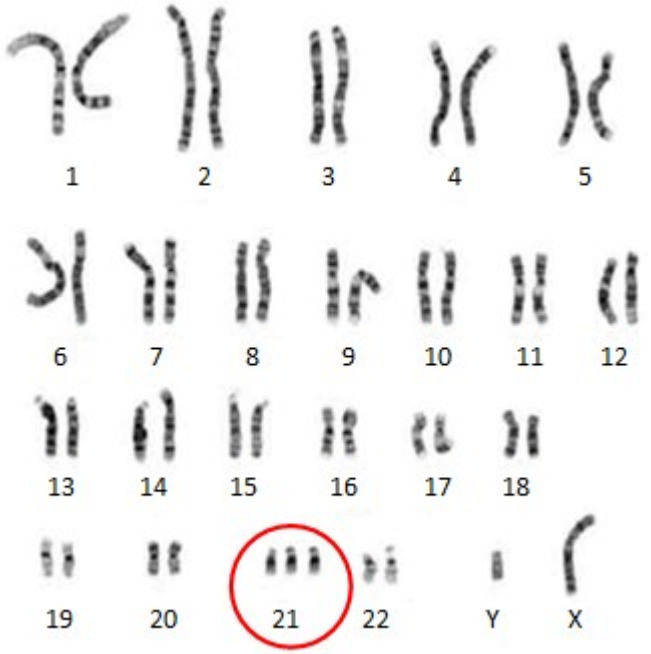
Figure 12.3 Types of chromosomal mutations

- In addition to these is *aneuploidy* where the chromosomes do not separate correctly during Meiosis I or II, resulting an abnormal chromosome number

Learning Aim: List and Describe Chromosomal Mutation Types

Example: Down Syndrome

- Trisomy 21:
  - Aneuploidy
  - 3 copies of Chromosome 21, rather than only 2.
  - Due to non-disjunction during meiosis
- Some features can also occur with partial trisomy:
  - Extra copy of some of chromo 21 gets translocated.
- Features:
  - Characteristic facial features
  - Low muscle tone
  - Intellectual disability (not in every case)
  - Heart defects (in some cases)





Learning Aim: List causes and features of Down Syndrome

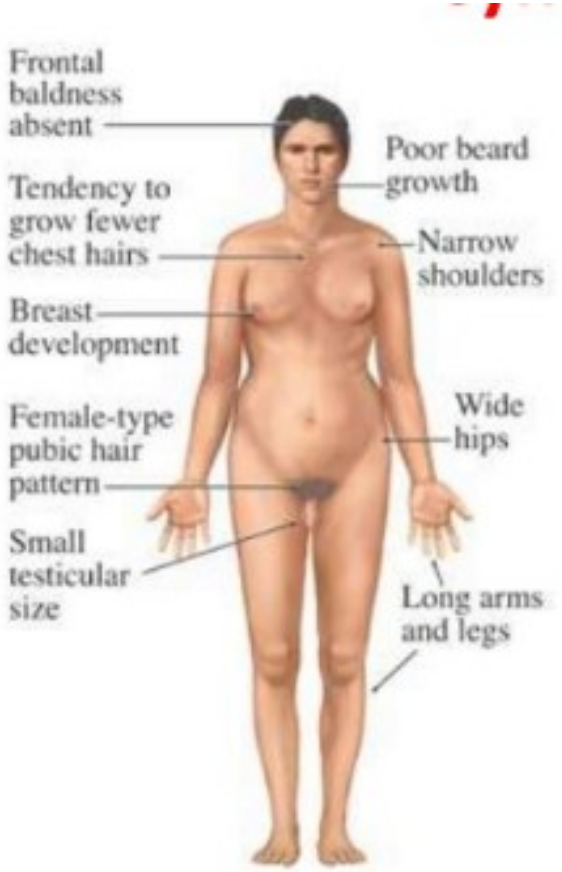
Example: Patau Syndrome

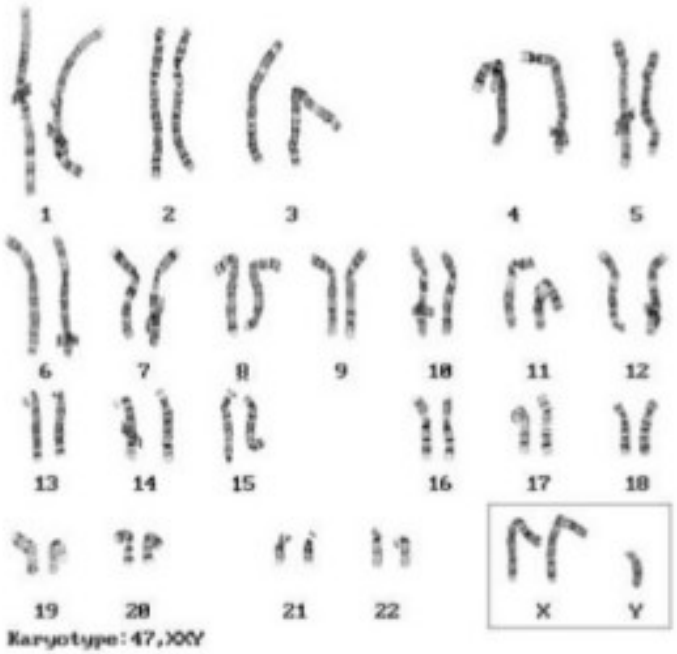
- Extra chromosome 13 (Trisomy 13)
- 1 in 5000 live births
- Affected babies usually die within a month of birth
- Range of serious physical defects to facial features, heart, body systems, brain.

Learning Aim: List causes and features of Patau Syndrome

Example: Klinefelter's Syndrome

- Affects Chromo 23 (sex chromosomes)
- XXY
- Male external characteristics
- Typical features start at puberty:
  - Small testes, no sperm produced
  - Enlargement of breasts
  - Less body hair
  - Sometimes intellectual disability

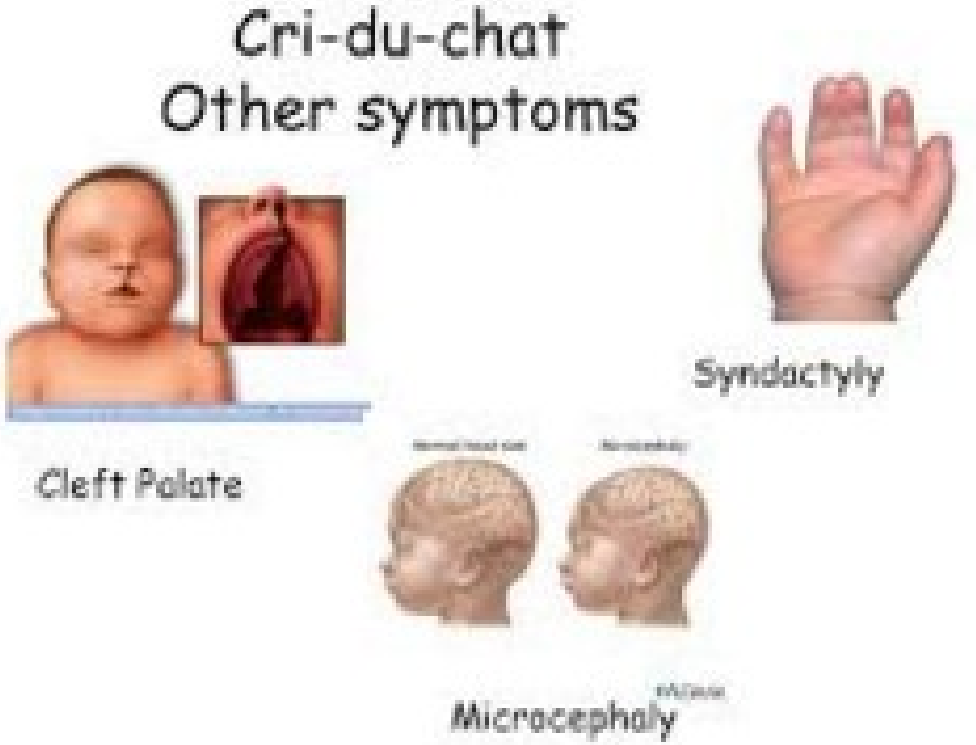




Learning Aim: List causes and features of Klinefelter's Syndrome

Example: Cri-du-chat syndrome

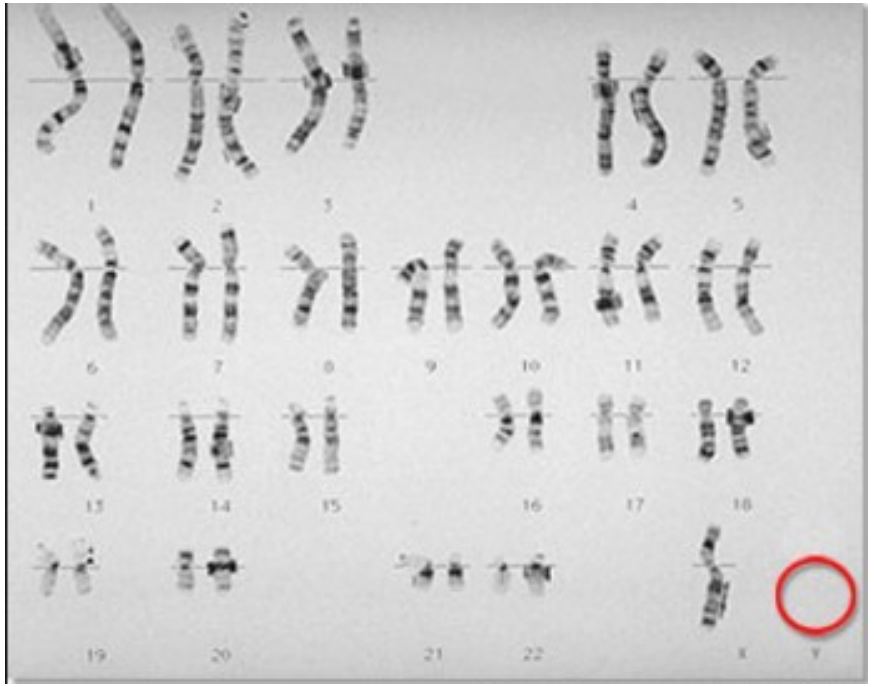
- Partial monosomy / partial chromosome deletion: part of Chromosome 5 is missing.
- Name because of unusual infant cry – problems with larynx and nervous system
- Features can include:
  - Cleft Palate
  - Fused/Webbed digits
  - Microcephaly
  - Intellectual Disability
  - Delayed growth and development

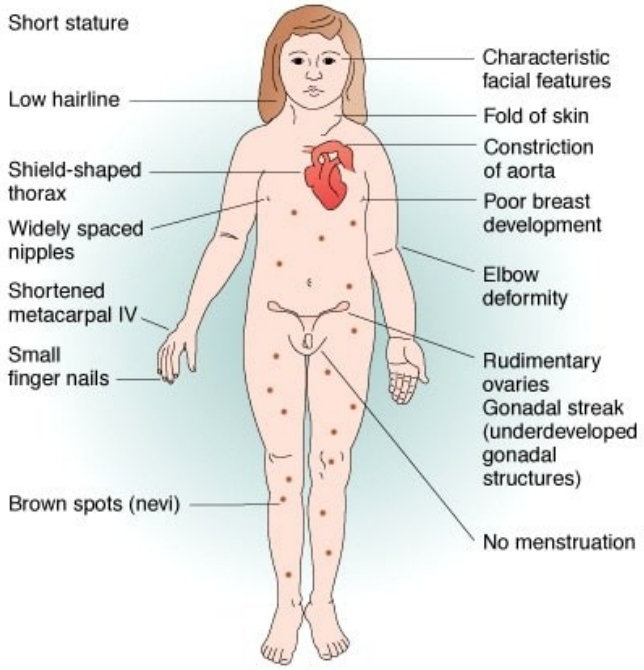


Learning Aim: List causes and features of Cri-du-chat Syndrome

Example: Turner's syndrome

- Monosomy: missing a sex chromosome.
  - Only 1 X, rather than XY or XX
- Physically appear female
  - Short stature
  - Lack secondary sex characteristics
  - Infertile

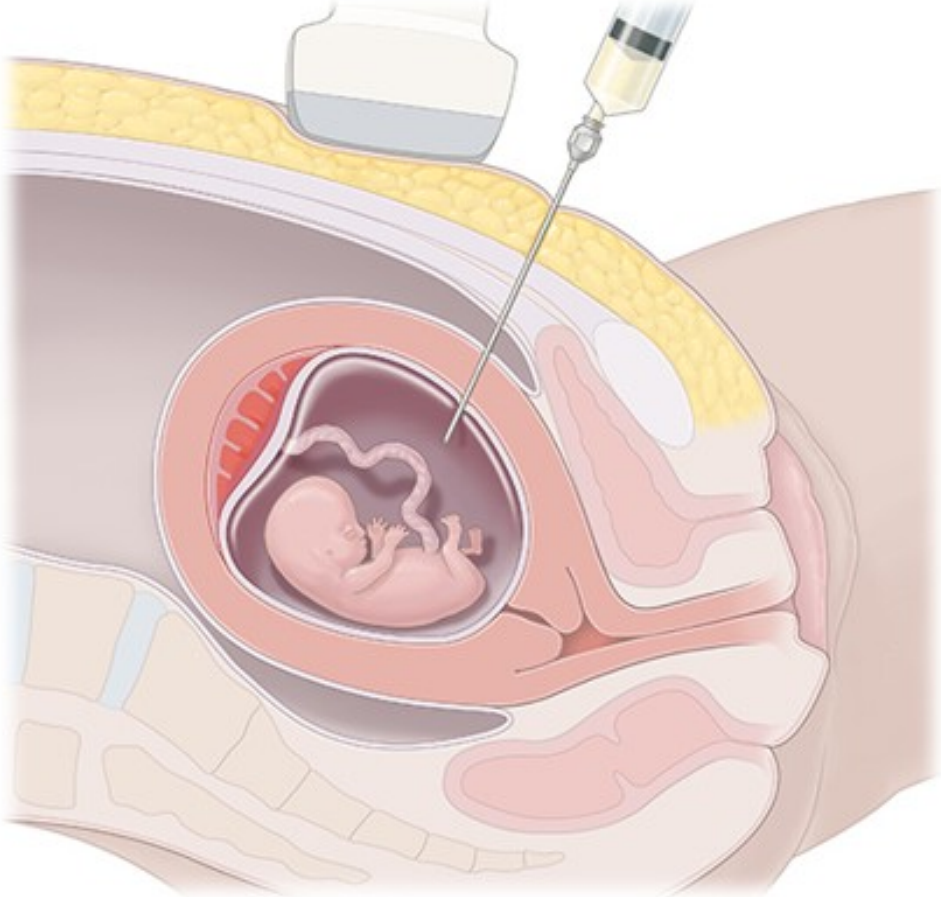




Learning Aim: List causes and features of Turner's Syndrome

In-utero diagnosis of Chromosomal Abnormalities

- By amniocentesis
- Blood sample analysis
- Placental sample analysis
- Can provide parents with information to assist with planning for child, or decision to terminate pregnancy,



Learning Aim: List methods for in utero detection of chromosomal abnormalities.